

Project Status Update

Learning Tire Degradation Dynamics in Formula 1

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1. Problem Understanding

The objective of this project is to model tire degradation in Formula 1 using machine learning. Initially, the problem was framed as predicting lap time degradation as a function of lap number and tire compound:

$$D(n) = f_{\theta}(n, c)$$

However, during experimentation, we observed that lap time is influenced by multiple competing factors:

- Tire degradation (desired signal)
- Driver behavior and skill
- Fuel load variation
- Race interactions (overtaking, defending)
- Track evolution and race conditions

Thus, the core challenge is not just prediction, but **isolating the degradation signal from noisy real-world data**. The problem can therefore be reframed as:

Estimating the component of lap time variation attributable to tire degradation under noisy and dynamic race conditions.

2. Approach

2.1 Initial Modeling

We began with a polynomial regression model to capture nonlinear degradation trends. However, initial results showed very low predictive performance due to significant noise in the dataset. This led to a key realization that the primary bottleneck is not model complexity, but data quality and noise.

2.2 Identifying Sources of Noise

A major source of noise arises from **race interactions**:

- When a driver is close to another car, they may:
 - Defend position (slower lap)

- Attempt overtakes (aggressive lap)
- These effects introduce abrupt deviations in lap times

Thus:

$$\text{LapTime} = \text{Tire Effect} + \text{Driver} + \text{Traffic/Interaction} + \text{Noise}$$

The interaction component introduces high-variance, non-systematic noise that obscures the degradation signal.

2.3 Clean Lap Filtering (Key Contribution)

To address this, we introduced a **clean lap filtering mechanism**.

We compute:

- Gap Ahead: time difference to the car in front
- Gap Behind: time difference to the car behind

A lap is considered “clean” if:

- Gap Ahead > threshold
- Gap Behind > threshold

This ensures:

- No immediate need for overtaking
- No defensive driving pressure
- Lap reflects “free air” driving conditions

2.4 Impact of Noise Reduction

Applying this filtering resulted in:

- Removal of anomalous laps caused by traffic
- Smoother degradation curves
- Significant improvement in model performance

Notably, the R^2 score improved from near-zero/negative values to approximately:

$$R^2 \approx 0.35 - 0.40$$

This demonstrates that race interaction effects are a major source of noise, and removing them significantly improves the learnability of tire degradation.

3. Data Status

We use lap level data from the FastF1 API across multiple 2024 races.

Preprocessing steps:

- Extracted relevant columns (LapTime, Compound, Stint, etc.)
- Converted lap times to numerical values
- Removed inaccurate laps (pit laps, safety car laps)
- Segmented stints and computed LapInStint
- Normalized lap times within each stint:

$$\Delta t = t_{\text{lap}} - t_{\text{first}}$$

Additional noise reduction:

- Gap based clean lap filtering
- Removal of extreme lap time spikes
- Restriction to stable stint windows

4. Initial Experiments

We trained regression models on both:

- Raw dataset
- Cleaned (filtered) dataset

Observations:

- Raw data produced poor predictions due to noise
- Cleaned data showed significantly improved trends
- Model predictions aligned better with expected degradation behavior

5. Future Directions: Noise Modeling and Reduction

While gap based filtering improves results, additional sources of noise remain(eg: Driver mistakes, race strategies). Future work will focus on identifying and mitigating these:

5.1 Additional Noise Sources

- Fuel load variation over race duration
- Driver specific pacing strategies
- Micro-events (mistakes, braking variability)

6. Planned Modeling Approaches

Once data quality is improved, we plan to explore:

- Polynomial regression (baseline)
- Feedforward neural networks (MLP)
(We will explore more models if we get more time, however we believe that noise modeling is more important over here)

The focus will be on:

- Capturing nonlinear degradation patterns
- Improving generalization across races and drivers

7. Progress and Next Steps

Completed:

- Data extraction and preprocessing pipeline
- Stint segmentation and normalization
- Identification of race interaction noise
- Development of clean lap filtering using gap metrics
- Demonstrated improvement in model performance after noise reduction

Next Steps:

- Refine clean lap identification metrics using gap consistency and stability measures
- Use machine learning models to analyze feature importance and identify which factors most strongly influence lap time
- Identify data points where non-tire-related features (e.g., race interactions, variability) dominate lap time behavior
- Remove or down-weight such data points to reduce noise and improve correlation between tire wear and lap time degradation
- Experiment with different model architectures on the refined dataset
- Compare degradation behavior across compounds using improved datasets

Conclusion

We demonstrate that race interactions introduce significant noise into lap time data, and that filtering these effects is essential for meaningful modeling of tire degradation. We believe that there is a clear nonlinear relationship between lap time in clean, free-air conditions and tire degradation. By isolating such laps and reducing external noise, we aim to better capture and model this relationship. We believe that the incorporation of such models would help teams strategize in a better way